Global principal-divisor and Picard-class proof boundary

Read-only formal dependency inventory

28 October 2026

Inspected isolated commit `f0c4308b62fa6a2621637d7fd02a13a148881ed4`. This packet adds no Lean theorem, experiment, target result, or cost ratio. Existing declarations are reported only with their stated hypotheses; the inventory did not rebuild all predecessors.

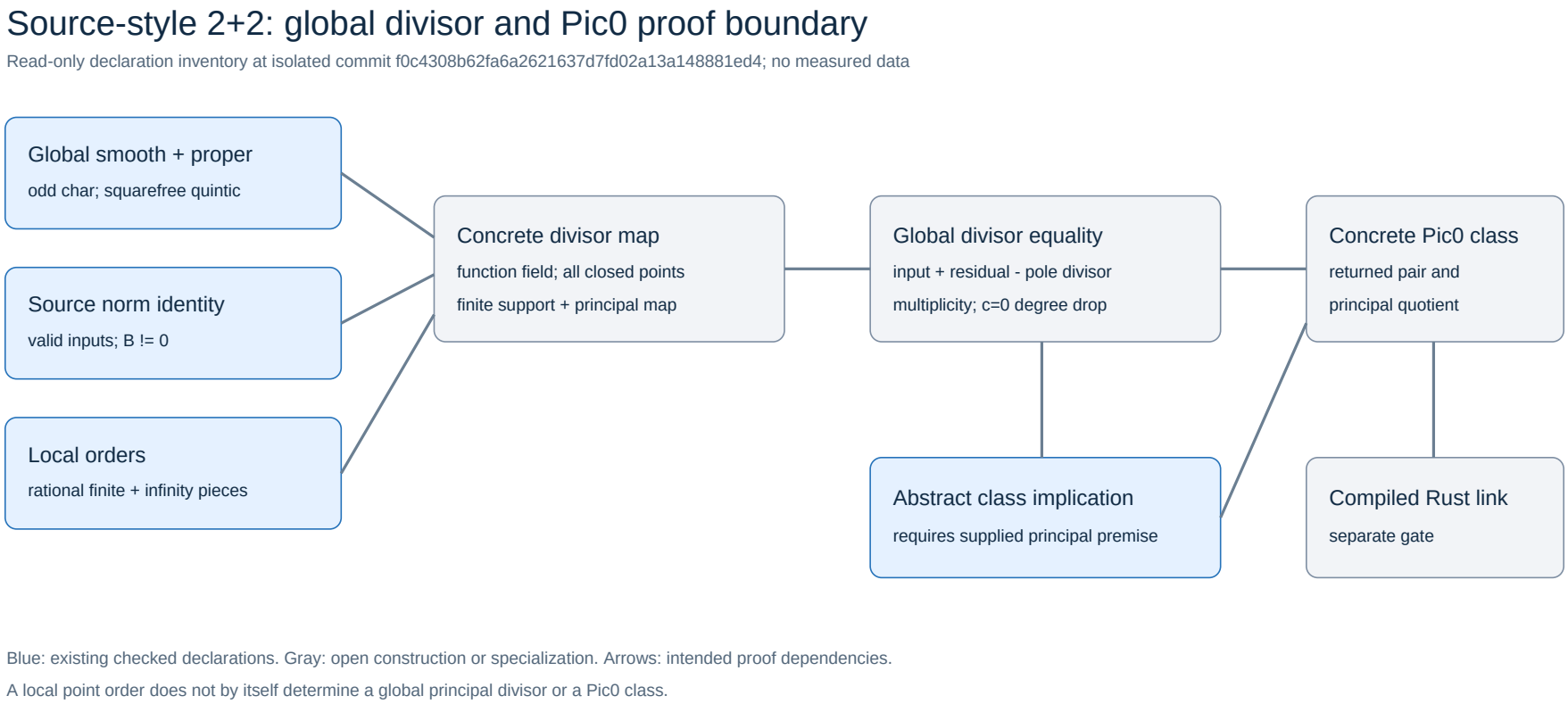
Checked components present

- `smooth`: actual weighted-Proj structure map smooth over a perfect field when $2 \neq 0$ and the quintic is squarefree. `weightedProj_proper_over_K`: properness over K. Full declaration names are in the CSV.
- `source_curve_eq`: source-style polynomial norm under valid input congruences and nonzero determinant. `returned_v_cleared`: output-ordinate polynomial congruence. Full declaration names are in the CSV.
- `WeightedProjNonbranchNormTransport`: source-style norm in an actual rational-point stalk; chosen-factor order with explicit separability, point, nonunit, $2y \neq 0$, $b \neq 0$, and norm premises. The separate branch theorem gives rational simple-branch order with an explicit support-root premise.
- `WeightedProjInfinityDVR.projectiveZ_order_two`: local infinity order. `Common22FractionPole`: pole-five/six implications only for a supplied valuation and cleared interpolant conditions.
- `ProjectiveLawRuntime.residual_represents_input`: abstract class implication only after a curve-specific principal-divisor interface and principal premise are supplied.

Open theorem boundary

Define the actual curve function $g = n/b - Y$ and effective divisors for the two input Mumford pairs and residual pair. A global proof must show that the local orders at *all closed points* (including nonrational points, branch overlap, repeated roots, and the $c = 0$ degree drop) and the infinity pole determine $\mathrm{div}(g)$. It must identify this divisor as principal, match the residual divisor with the returned pair and its involution, and construct the concrete degree-zero class map. The schematic equation $\mathrm{div}(g) = D_1 + D_2 + Z_{\mathrm{res}} - m\infty$ is a proposed target, not a checked theorem; m needs a proof by degree branch. A ring Picard group in pinned Mathlib is not already a curve-specific divisor-class construction. The abstract group implication alone cannot discharge the principal premise.
Sources: repository declarations listed in the accompanying Markdown/CSV; Stacks Project Tags 02RS (principal divisors) and 03RN (Picard groups of curves). These sources inform obligations; they do not certify the missing Lean bridge.

Proof dependencies



Blue boxes are existing checked declarations; gray boxes are open construction or specialization gates. Arrows show intended dependencies, not measured performance.